DITTERT CONJECTURE IN EVERY DIMENSION VIA UNIFORM BOUNDARY RIGIDITY

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Editorial note. Complete original paper106–1954, all nonadjacent rational certificate appendices, independent proof routes and embedded bibliography retained. Authors retain responsibility for mathematics; no added assumptions, proof repair, author verification or independent computation/certification.

Editorial note. Original lines1823–1824 AI-assisted discovery/human-verification disclosure retained exactly: TARS assisted exploratory derivations with autonomous mathematical search; X.Dai states proof reconstruction, omitted details, literature/manuscript work and manual verification. This does not explicitly bind discovery of the selected core proof to AI; under current project instruction no whole-paper AI-core inference, AI-free claim or dependency-chain investigation is made.

Editorial note. Count only thm:dittert in every dimension. Boundary exclusion, spectral dimensions7–13 and prescribed-zero dimensions14–15 are bundled original routes/context, not separate counts.

Allowed standard prerequisites (editorial).

- Matrix permanents, positive-support maximizers, one-sided KKT conditions and rational polynomial positivity. Original nonnegative-matrix normalization, permanent and row/column products, precise author-used Hwang/Cheon-Wanless statements with original references, and all author printed exact rational certificates retained.. Native boundary reduction, support-stratum and uniform-multiplicity arguments are collected in full; no independent certificate computation or proof reconstruction is performed.

1. Introduction

For an $n \times n$ matrix $A = (a_{ij})$, its permanent is

$$\operatorname{per} A = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

The permanent is one of the central nonlinear functionals of combinatorial matrix theory; standard references include Minc's monograph and Brualdi–Ryser's treatment [17, 2]. Let

$$\mathcal{K}_n = \left\{ A \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j} a_{ij} = n \right\}.$$

For $A \in \mathcal{K}_n$, write $r_i = \sum_j a_{ij}$ and $c_j = \sum_i a_{ij}$, and set

$$R(A) = \prod_{i=1}^{n} r_i, \qquad C(A) = \prod_{j=1}^{n} c_j, \qquad \Phi(A) = R(A) + C(A) - \operatorname{per} A.$$

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If $\Omega_n \subset \mathcal{K}_n$ denotes the Birkhoff polytope, then $R(A) = C(A) = 1$ on Ω_n. Consequently, the inequality

$$\Phi(A) \le 2 - \frac{n!}{n^n}$$

reduces on Ω_n to the van der Waerden lower bound $\operatorname{per} A \ge n!/n^n$, proved by Egorychev and Falikman; see, for example, [9, 17]. Dittert's problem asks whether the same extremal matrix remains unique when the row and column sums are allowed to vary over the larger simplex $\mathcal{K}_n$.

Conjecture 1.1 (Dittert)**.** *For every $n \ge 2$ and every $A \in \mathcal{K}_n$,*

$$\Phi(A) \le 2 - \frac{n!}{n^n},$$

with equality if and only if $A = U_n := J_n/n$.

The conjecture appears as Conjecture 28 in Minc's survey [18]; its subsequent development is reviewed in [5]. Sinkhorn proved the case $n = 2$, as well as positivity of the permanent at a maximizing matrix [22]. Hwang established the case $n = 3$, proved that every positive maximizer is U_n, and derived the entrywise multiplier relations that underlie essentially all later boundary analyses [13, 14]. Hwang–Sohn–Kim studied additional classes of matrices adapted to the Dittert functional [12], while Cheon–Hwang introduced a family of sub-Dittert functionals [4]. Further structural and sufficient conditions were obtained through monotonicity, permanental-minor, and averaging arguments [3, 8].

Two later papers sharpened the geometric picture. Cheon–Wanless showed that a maximizing matrix is fully indecomposable, excluded zero sets that form a single rectangular block, and proved strong quantitative restrictions on a near maximizer [7]. They also gave an equivalent weighted-semimatching formulation of the conjecture [6]. Udayan–Somasundaram subsequently proved that an order-four maximizer is fully indecomposable [23]; this leaves the genuinely boundary, fully indecomposable support strata untreated. Most recently, Pang proved the conjecture for $n \ge 17$, and Kafidov treated $n = 16$, in two preprints based on increasingly sharp scaling estimates [21, 15]. Thus the remaining difficulty is concentrated in small and intermediate dimensions, where the asymptotic scaling margins are insufficient.

The present paper closes the problem by isolating a uniform boundary mechanism. Hwang's positive-support theorem reduces the conjecture to the exclusion of boundary maximizers. The Cheon–Wanless zero-set theorem then forces a hypothetical boundary maximizer to contain two zero entries in distinct rows and distinct columns. After moving these entries to $(1, 1)$ and $(2, 2)$, an attained-stratum averaging argument produces a maximizer in which the remaining $n - 2$ rows and columns are equal. The resulting matrix has seven positive parameters and a single multiplicity $m = n - 2$.

The proof then separates into two regimes. In order four, the repeated block has multiplicity two and the boundary contains several additional support degenerations. We first enumerate the admissible fully indecomposable support types up to row permutations, column permutations, and transpose. On the two-zero stratum, three odd perturbation variables reduce the stationarity equations to a quartic form. The sum of the two missing-entry multiplier defects has an explicit cofactor expression. Exact Hessian inequalities and a tensor-product Bernstein certificate force this sum to be positive, contradicting the one-sided KKT conditions.

For $n \ge 5$, the same odd-variable expansion retains the multiplicity $m = n - 2$ symbolically. The local maximality equations produce two quadratic coefficients α and δ, while the missing-entry cofactor identity reduces the obstruction to two uniform inequalities. Their numerators are expanded coefficientwise in the normalized

variables and in the multiplicity m. Positivity follows from elementary binomial estimates valid for every $m \geq 3$. This is the decisive step: the all-dimensional argument is symbolic, rather than an interpolation from finitely many computer-checked cases.

Our main result is the following.

Theorem 1.2. *For every integer $n \geq 2$ and every nonnegative $n \times n$ matrix A whose entries sum to n,*

$$\prod_{i=1}^{n} r_i + \prod_{j=1}^{n} c_j - \operatorname{per} A \leq 2 - \frac{n!}{n^n}.$$

Equality holds if and only if $A = J_n/n$.

The load-bearing boundary statement can be separated from the known positive-support result.

Theorem 1.3 (Boundary exclusion)**.** *For every $n \geq 4$, no global maximizer of Φ on $\mathcal{K}_n$ has a zero entry.*

The order-four part of Theorem 1.3 is proved in Section 5; the uniform $n \geq 5$ part is proved in Section 6. Together with Hwang's positive-support theorem and the known cases $n = 2, 3$, this immediately yields Theorem 1.2.

We retain two independent proof routes because they clarify different parts of the geometry. Section 3 converts Hwang's multiplier equation into an eigenvalue equation for a reversible Markov chain and excludes dimensions 7 through 13 by exact cut and spectral estimates. Section 4 combines two prescribed zeroes with exact minimum-permanent calculations to treat dimensions 14 and 15. These arguments are not needed after the uniform boundary theorem, but they provide independent checks and show precisely where purely marginal scaling estimates cease to be sharp.

All finite calculations are proof certificates rather than floating-point experiments. Sturm sequences are used for exact univariate root counts and rational isolating intervals; standard background is given in [1]. Polynomial positivity on rectangles is certified in the Bernstein basis, whose positivity and convex-hull properties are reviewed in [10]. Every rational threshold, support orbit, coefficient table, and verification routine needed by the proof is printed in the appendices.

The paper follows the logical structure of the argument. Section 2 records the extremal reductions and the two-zero boundary consequence. Sections 3 and 4 give the independent finite-range proofs. Section 5 treats the order-four boundary, and Section 6 proves the symbolic multiplicity theorem for all $n \geq 5$. Section 7 combines the cases. The appendices contain the exact rational spectral checks, Sturm data, capacity countermodels, complete order-four support list, Bernstein table, and symbolic bookkeeping protocol.

2. Preliminaries and the boundary reduction

Throughout the paper, $n \geq 2$, $A \in \mathcal{K}_n$, and

$$R = \prod_i r_i, \qquad C = \prod_j c_j, \qquad P = \operatorname{per} A, \qquad \gamma_n = \frac{n!}{n^n}.$$

The matrix $U_n = J_n/n$ satisfies $R(U_n) = C(U_n) = 1$, $\operatorname{per} U_n = \gamma_n$, and hence $\Phi(U_n) = 2 - \gamma_n$.

2.1. **The elementary extremal reduction.**

Lemma 2.1. *For every $A \in \mathcal{K}_n$, one has $R \le 1$ and $C \le 1$. If a global maximizer A satisfies $P = \gamma_n$, then $A = U_n$.*

Proof. The row sums and column sums are nonnegative and each family has total sum n, so both product bounds follow from AM–GM. If A is a global maximizer and $P = \gamma_n$, then

$$R + C - \gamma_n = \Phi(A) \ge \Phi(U_n) = 2 - \gamma_n.$$

Thus $R = C = 1$, and equality in AM–GM makes every row sum and every column sum equal to one. Hence $A \in \Omega_n$. Equality in the van der Waerden theorem then gives $A = U_n$ [9]. □

The following two results are the structural input from the previous literature.

Theorem 2.2 (Hwang)**.** *A positive global maximizer of Φ on $\mathcal{K}_n$ is U_n. Moreover, at a maximizing matrix the directional derivatives at positive entries agree, while the derivative at a zero entry satisfies the corresponding one-sided inequality.*

Proof. These are the positive-support theorem and multiplier conditions proved in [13, 14]; the precise forms used below are also recorded in [7, Lemmas 3.5–3.6]. □

Theorem 2.3 (Cheon–Wanless)**.** *Every global maximizer of Φ is fully indecomposable. In addition, the complete zero set of a maximizing matrix cannot be a single rectangular block.*

Proof. See [7, Theorems 3.2 and 3.7]. □

2.2. **Two independent zeroes.** Call two positions independent if they lie in distinct rows and distinct columns.

Corollary 2.4. *Every boundary maximizer contains two independent zero entries.*

Proof. View the zero positions as edges of a bipartite graph on the row and column vertices. If no two zero edges are independent, then every pair of edges intersects. A pairwise-intersecting family of edges in a bipartite graph is a star, so all zeroes lie in one row or all lie in one column. The complete zero set is then a rectangular block, contrary to Theorem 2.3. □

2.3. **Exact-certificate convention.** Whenever a dimension-dependent inequality is verified computationally, all inputs are rational and all comparisons are performed after cross-multiplication in $\mathbb{Z}$. Root counts use exact Sturm variation, interval evaluations use rational Horner arithmetic, and positive definiteness is certified by rational LDL^T pivots. Decimal values are included only for orientation and never carry a proof.

3. A spectral exclusion in dimensions 7 through 13

Theorem 3.1. *The Dittert conjecture holds for $7 \le n \le 13$.*

Proof. Let A be a global maximizer and put

$$R = \prod_i r_i, \qquad C = \prod_j c_j, \qquad P = \operatorname{per} A.$$

Write $P = \gamma_n - \delta$. Lemma 2.1 and maximality give $0 \le \delta \le \gamma_n$, with $\delta = 0$ only at U_n. Theorem 2.2 reduces the proof to a boundary maximizer, so throughout this section $0 < \delta \le \gamma_n$.

Cheon–Wanless prove that such a maximizer is fully indecomposable [7, Theorem 3.2]. Hwang's multiplier equation [7, Lemmas 3.5–3.6], multiplied by a_{ij} and summed by rows and columns, gives, with $u = r - \mathbf{1},\ v = c - \mathbf{1}$,

$$(R - P)u = -C\,AD_c^{-1}v, \qquad (C - P)v = -R\,A^T D_r^{-1}u. \tag{6}$$

If $u = 0$, Hwang's row-stochastic-maximizer theorem already gives $A = U_n$. Otherwise

$$AD_c^{-1}A^T D_r^{-1}u = \theta u, \qquad \theta = \frac{(R - P)(C - P)}{RC}. \tag{7}$$

The transpose

$$\widehat{K} = D_r^{-1}AD_c^{-1}A^T$$

is a reversible row-stochastic chain with stationary probabilities $\pi_i = r_i/n$. Full indecomposability makes it irreducible. This is the standard spectral framework for finite reversible chains; see [16]. Detailed balance turns $D_\pi^{-1}u$ into a nonconstant right eigenfunction with eigenvalue θ. Since $R, C \le 1$ and $\Phi(A) \ge 2 - \gamma_n$,

$$1 - \theta \le 2\gamma_n - \gamma_n^2. \tag{8}$$

Put

$$a_n = \sqrt{\frac{n\gamma_n}{2(1 - \gamma_n)}}.$$

The subset-sum estimate in the proof of Kafidov's Lemma 3.1 gives, for all row and column sets,

$$|r(I) - |I|| \le a_n, \qquad |c(J) - |J|| \le a_n. \tag{9}$$

Order the values of the eigenfunction, put $d_j = f_{j+1} - f_j$, and let S_j be the first j states. Write $\ell_j = \min(j, n - j)$ and $Q_j = Q(S_j, S_j^c)$. If every such cut has conductance at least H_{ℓ_j}, then reversibility, (9), and $(f_s - f_i)^2 \ge \sum_{j=i}^{s-1} d_j^2$ give

$$\begin{aligned}(1 - \theta)\,\mathrm{Var}_\pi(f) &\ge \sum_{j=1}^{n-1} d_j^2 Q_j \\ &\ge \sum_{j=1}^{n-1} d_j^2 H_{\ell_j}\,\frac{\ell_j - a_n}{n}.\end{aligned}$$

Cauchy–Schwarz and $\mathrm{Var}_\pi(f) \le (\max f - \min f)^2/4$ imply

$$1 - \theta \ge \frac{4}{\displaystyle\sum_{j=1}^{n-1} n/[H_{\ell_j}(\ell_j - a_n)]}. \tag{10}$$

Suppose instead that a cut S, chosen on its stationary smaller side, has cardinality ℓ and conductance $h < H_\ell$. By (9) and $a_n < 1/2$, this is the smaller cardinality of the two sides (with either choice allowed for the balanced cut). Set

$$q_j = \frac{\sum_{i \in S} a_{ij}}{c_j}, \qquad T = \{j : q_j \ge 1/2\}.$$

The mass E in the two off-diagonal rectangles satisfies the exact flow bound

$$E \le 2\sum_j c_j q_j (1 - q_j) = 2h\,r(S) < 2H_\ell(\ell + a_n). \tag{11}$$

Put $\eta_\ell = 2H_\ell(\ell + a_n) + 3a_n$. Equations (9)–(11) first give

$$\big||T| - \ell\big| \le E + 2a_n < \eta_\ell - a_n < 1,$$

so $|T| = \ell$. For $I \subseteq S, J \subseteq T$, inclusion–exclusion inside the diagonal block gives

$$s(A[I,J]) \geq |I| + |J| - \ell - E - 3a_n,$$

and the same holds in the complementary diagonal block. The doubly-superstochastic criterion and van der Waerden therefore give

$$P \geq \gamma_\ell \gamma_{n-\ell}(1-\eta_\ell)^n. \tag{12}$$

It remains only to choose the thresholds. All numerical quantities below are rational. When a reduced margin has an unwieldy numerator and denominator, we print a shorter strict rational lower bound sufficient for the comparison; no decimal approximation is proof-bearing. The following entries are exact rationals:

n	a_n^+	$10^4(H_1,\dots,H_{\lfloor n/2\rfloor})$
12	$180/10^4$	$(100,90,75,60,50,40)$
13	$116/10^4$	$(160,120,95,75,60,50)$.

The exact-arithmetic procedure in Appendix A checks over $\mathbb{Q}$ that $a_n < a_n^+$, that for every listed ℓ

$$\gamma_\ell \gamma_{n-\ell}\big(1 - 2H_\ell(\ell + a_n^+) - 3a_n^+\big)^n > \gamma_n,$$

and that the right side of (10), weakened by replacing a_n with a_n^+, exceeds $2\gamma_n - \gamma_n^2$. Thus a low-conductance cut contradicts $P \leq \gamma_n$ through (12), while the absence of such a cut contradicts (8) through (10). This excludes boundary maximizers for $n = 12, 13$.

For $n = 11$, replace the uniform bound (9) by an exact size-dependent one. If a set of k row or column sums has total $k + \varepsilon$, AM–GM gives

$$\prod_i r_i \text{ or } \prod_j c_j \leq F_k(\varepsilon) := \left(1 + \frac{\varepsilon}{k}\right)^k \left(1 - \frac{\varepsilon}{11-k}\right)^{11-k}.$$

The function decreases away from zero on either side. Direct rational evaluation, recorded in Appendix A, verifies $F_k(\pm b_k) < 1 - \gamma_{11}$, and hence $|\varepsilon| < b_k$, for the exact rationals

$$10^6(b_1,\dots,b_{10}) = (16030, 21460, 24755, 26720, 27640, 27640, 26720, 24755, 21460, 16030).$$

In (10), replace a_n for a cut of size ℓ by b_ℓ. For a diagonal block of order m, the Hall marginal loss becomes

$$B_m = b_m + \max_{\substack{1 \leq i,j \leq m \\ i+j > m}} (b_i + b_j).$$

Thus (12) holds with

$$\eta_\ell = 2H_\ell(\ell + b_\ell) + \max(B_\ell, B_{11-\ell}).$$

The same appendix checks the preliminary block-order inequality and all block and spectral comparisons for

$$10^4(H_1,\dots,H_5) = (45, 55, 45, 38, 32).$$

Its smallest exact margins in the marginal, block, and spectral checks are strictly greater than the rational numbers

$$\frac{1}{40000000}, \qquad \frac{1}{250000}, \qquad \frac{3}{50000},$$

respectively. Hence dimension 11 is excluded as well.

For $n = 10$, retain additionally the shared deficit budget. Write $R = 1 - \rho, C = 1 - \sigma$; then $\rho + \sigma \le \delta \le \gamma_{10}$. Preliminary exact AM–GM bounds first put every subset-sum error inside $\frac{43}{1000}$. On that interval,

$$(-\log F_k)'' \ge \frac{k}{(k + \frac{43}{1000})^2} + \frac{10 - k}{(10 - k + \frac{43}{1000})^2}.$$

Consequently a row-set error is at most $c_k\sqrt{\rho}$, and a column-set error at most $c_k\sqrt{\sigma}$, for the rational upper bounds

$$10^3(c_1, \dots, c_9) = (1395, 1822, 2075, 2212, 2256, 2212, 2075, 1822, 1395).$$

Cauchy–Schwarz with $\rho + \sigma \le \gamma_{10}$ now bounds the two row-error terms and one column-error term in each block Hall inequality jointly. The exact rational data are

ℓ	1	2	3	4	5
$10^4 e_\ell$	266	348	396	422	430
$10^4 L_\ell$	818	889	931	955	962
$10^4 H_\ell$	41	57	48	40	33,

where $e_\ell > c_\ell\sqrt{\gamma_{10}}$, L_ℓ bounds every joint Hall loss, and H_ℓ is the cut threshold. Thus the low-conductance alternative gives (12) with $\eta_\ell = 2H_\ell(\ell + e_\ell) + L_\ell$.

For the other alternative put $w_j = H_{\ell_j}(\ell_j - e_{\ell_j})/10$. Weighted Cauchy–Schwarz along the ordered path gives

$$\mathrm{Var}_\pi(f) \le \left(\frac{1 + e_1}{10}\right)^2 \max_h \left(\sum_{i \le h < r} \sum_{q=i}^{r-1} \frac{1}{w_q}\right) \sum_{j=1}^{9} w_j d_j^2.$$

Hence the reciprocal of the first two factors is a lower bound for $1 - \theta$. Appendix A verifies all defining rational inequalities; its smallest marginal, block, and spectral margins are strictly greater than

$$\frac{19}{1000000000}, \qquad \frac{9}{10000000}, \qquad \frac{1}{10000},$$

respectively. These contradict (8) in both alternatives, excluding dimension 10.

For $n = 9$, retain the same deficit budget and sharpen the block estimate itself. Exact AM–GM comparisons first bound every k-subset error by

$$10^4(b_1, \dots, b_8) = (414, 544, 615, 647, 647, 615, 544, 414).$$

In particular all errors have absolute value less than $\frac{13}{100}$. The lower bound

$$(-\log F_k)'' \ge \frac{k}{(k + \frac{13}{100})^2} + \frac{9 - k}{(9 - k + \frac{13}{100})^2}$$

therefore proves that row-set errors are at most $c_k\sqrt{\rho}$ and column-set errors at most $c_k\sqrt{\sigma}$, where

$$10^3(c_1, \dots, c_8) = (1488, 1861, 2074, 2172, 2172, 2074, 1861, 1488).$$

Exact rational comparison also gives the absolute bounds

$$10^4(e_1, \dots, e_8) = (456, 570, 635, 665, 665, 635, 570, 456), \qquad e_k > c_k\sqrt{\gamma_9}.$$

Let S be a low-conductance cut on its stationary smaller side, $\ell = |S| \le 4$, and define T by the same column threshold as in (11). The off-diagonal mass satisfies

$$E < 2H_\ell(\ell + e_\ell).$$

The preliminary b_k bounds force $|T| = \ell$. For either diagonal block, let its row and column sets be $\mathcal{U}, \mathcal{V}$, of order m. If $I \subseteq \mathcal{U},\ J \subseteq \mathcal{V}$, then

$$s(A[I, J]) \ge c(J) - r(\mathcal{U} \setminus I) - E,$$
$$s(A[I, J]) \ge r(I) - c(\mathcal{V} \setminus J) - E.$$

Writing $i = |I|,\ j = |J|$, the two losses below $i + j - m$ are at most

$$c_{m-i}\sqrt{\rho} + c_j\sqrt{\sigma}, \qquad c_i\sqrt{\rho} + c_{m-j}\sqrt{\sigma}.$$

Consequently their minimum is at most L_m provided

$$L_m^2 > \gamma_9 \max_{i+j>m} \min\{c_{m-i}^2 + c_j^2, c_i^2 + c_{m-j}^2\}, \quad c_0 = 0.$$

All of these strict comparisons hold for the exact rational data

ℓ	$10^4 L_\ell$	$10^4 L_{9-\ell}$	$10^4 H_\ell$
1	456	920	55
2	570	920	90
3	730	876	80
4	782	853	65.

Applying the doubly-superstochastic criterion separately to the two blocks gives, for every ℓ,

$$P \geq \gamma_\ell \gamma_{9-\ell}(1 - E - L_\ell)^\ell (1 - E - L_{9-\ell})^{9-\ell} > \gamma_9,$$

a contradiction.

Finally suppose no low-conductance sweep cut exists and put

$$w_j = H_{\ell_j} \frac{\ell_j - e_{\ell_j}}{9}.$$

The eigenfunction energy is at least $\sum_j w_j d_j^2$. If

$$M_{qr} = \min(q, r)(9 - \max(q, r)),$$

then its uniform variance is $9^{-2} d^T M d$. Exact rational LDL^T elimination as in Appendix A proves that

$$\operatorname{diag}(w_1, \ldots, w_8) - \frac{M}{500 \cdot 9^2}$$

is positive definite. Since $\pi_i \leq (1 + e_1)/9$, stationary variance is at most $1 + e_1$ times uniform variance. Hence

$$1 - \theta \geq \frac{1/500}{1 + e_1} > 2\gamma_9 - \gamma_9^2,$$

contradicting (8). The smallest exact marginal, Hall, block-permanent, path-pivot, and spectral margins are respectively greater than

$$\frac{1}{500000}, \quad \frac{3}{1000000}, \quad \frac{1}{125000}, \quad \frac{1}{5000}, \quad \frac{1}{25000}.$$

This excludes dimension 9.

For $n = 8$, the decisive refinement is to retain the product deficit in the final comparison. Put $d = \rho + \sigma$. Since $d \leq \delta$,

$$P = \gamma_8 - \delta \leq \gamma_8 - d. \tag{13}$$

The exact preliminary bounds and error coefficients are

$$10^4(b_1, \ldots, b_7) = (662, 859, 955, 982, 955, 859, 662),$$
$$10^3(c_1, \ldots, c_7) = (1555, 1877, 2049, 2103, 2049, 1877, 1555),$$
$$10^4(e_1, \ldots, e_7) = (763, 921, 1005, 1032, 1005, 921, 763).$$

They certify $|\varepsilon| < b_k < 1/5$, subset errors bounded by $c_k\sqrt{\rho}$ or $c_k\sqrt{\sigma}$, and $e_k > c_k\sqrt{\gamma_8}$.

The direct complementary-subset estimates used in dimension 9 now give Hall losses $K_m\sqrt{d}$. Exact Cauchy–Schwarz comparisons hold for

ℓ	10^3K_ℓ	$10^3K_{8-\ell}$	10^4H_ℓ
1	1556	2937	300
2	1878	2819	210
3	2438	2779	150
4	2573	2573	120.

For a cut below H_ℓ, put $x=\sqrt{d}$. Its off-diagonal mass is less than $2H_\ell(\ell+c_\ell x)$, and separate scaling of the two diagonal blocks gives

$$\begin{aligned} P \ge \gamma_\ell\gamma_{8-\ell}\,&(1-2H_\ell\ell-(2H_\ell c_\ell+K_\ell)x)^{\ell} \\ &\times(1-2H_\ell\ell-(2H_\ell c_\ell+K_{8-\ell})x)^{8-\ell}. \end{aligned} \tag{14}$$

For each ℓ, the right side of (14), minus γ_8-x^2, is an exact degree-eight polynomial. Appendix A verifies its positivity on $0\le x\le 491/10^4$, an interval containing $[0,\sqrt{\gamma_8}]$, by rational interval Horner evaluation on 16 equal subintervals. This contradicts (13).

If no low-conductance cut exists, use

$$w_j=H_{\ell_j}\frac{\ell_j-e_{\ell_j}}{8}.$$

Exact LDL^T elimination on the uniform-variance increment matrix proves that the weighted path energy is at least $11/2000$ times uniform variance. Since stationary variance is at most $1+e_1$ times uniform variance,

$$1-\theta\ge\frac{11/2000}{1+e_1}>2\gamma_8-\gamma_8^2,$$

contradicting (8). The smallest exact marginal, Hall, polynomial-interval, path-pivot, and spectral margins are respectively greater than

$$\frac{1}{200000},\quad\frac{1}{1000},\quad\frac{1}{20000},\quad\frac{1}{500},\quad\frac{3}{10000}.$$

This excludes dimension 8.

It remains to handle $n=7$. Keep $d=\rho+\sigma$ and $x=\sqrt{d}$. The same maximality inequality gives $P\le\gamma_7-d$, but now retain it also in the eigenvalue estimate. From (7),

$$\begin{aligned} \theta &\ge \frac{(1-\gamma_7+\sigma)(1-\gamma_7+\rho)}{(1-\rho)(1-\sigma)}, \\ 1-\theta &\le \frac{(2-\gamma_7)(\gamma_7-d)}{1-d}. \end{aligned} \tag{15}$$

The exact marginal and Hall coefficients are

$$\begin{aligned} 10^3(b_1,\dots,b_6)&=(106,135,146,146,135,106),\\ 10^3(c_1,\dots,c_6)&=(1484,1801,1941,1941,1801,1484),\\ 10^3(K_1,\dots,K_6)&=(1485,1802,2334,2444,2649,2649). \end{aligned} \tag{16}$$

Exact rational evaluation verifies $F_{7,k}(\pm b_k)<1-\gamma_7$, the Hessian comparisons on $|\varepsilon|<3/20$, and every direct Hall comparison.

We need one additional elementary estimate. For a row cut S, set

$$q_j=\frac{s(A[S,\{j\}])}{c_j},\quad T=\{j:q_j\ge1/2\},\quad Q=\sum_j c_jq_j(1-q_j).$$

Write

$$X=\sum_{j\in T}c_j(1-q_j),\qquad Y=\sum_{j\notin T}c_jq_j,\qquad E=X+Y.$$

Then $Y-X=r(S)-c(T)=:\Delta$. Once the preliminary estimate $E\le 2Q$ forces $|T|=|S|=\ell$, (16) gives

$$|\Delta|\le\sqrt{2}c_\ell x.$$

Every column sum is at least $c_{\min}=1-c_1x$. Consequently

$$\begin{aligned}Q&=E-\sum_{j\in T}c_j(1-q_j)^2-\sum_{j\notin T}c_jq_j^2\\&\ge E-\frac{X^2+Y^2}{c_{\min}}=E-\frac{E^2+\Delta^2}{2c_{\min}}.\end{aligned}\tag{17}$$

This is the point at which the dimension-eight factor-two estimate loses too much.

Take

$$10^4(H_1,H_2,H_3)=(502,373,276).$$

If a sweep cut has conductance below H_ℓ, then $Q<H_\ell(\ell+c_\ell x)$. The finite procedure in Appendix A partitions $0\le x\le 783/10^4$ into 4096 rational intervals. On each it chooses a multiple B of $\frac{1}{10^8}$ and verifies

$$\begin{aligned}&2(1-c_1x)\big(B-H_\ell(\ell+c_\ell x)\big)-B^2-2c_\ell^2x^2>0,\\&\gamma_\ell\gamma_{7-\ell}(1-B-K_\ell x)^\ell(1-B-K_{7-\ell}x)^{7-\ell}-\gamma_7+x^2>0.\end{aligned}\tag{18}$$

On the second line the two parenthesized powers are factors. The same exact calculation also checks $E<1-c_1x$, so (17) is used on its increasing branch. The first inequality gives $E<B$; separate block scaling and the second give $P>\gamma_7-d$, a contradiction.

If no sweep cut is small, set

$$w_j=H_{\ell_j}\frac{\ell_j-c_{\ell_j}x}{7}.$$

Exact rational LDL^T pivots give a uniform path gap $135/10^4$. Stationary variance is at most $1+c_1x$ times uniform variance, and exact interval arithmetic verifies

$$\frac{135}{10^4}(1-x^2)-(1+c_1x)(2-\gamma_7)(\gamma_7-x^2)>0.$$

This contradicts (15). The smallest exact marginal, Hall, leakage, block, path-pivot, and spectral margins are respectively greater than

$$\frac{3}{100000},\quad\frac{1}{1000},\quad\frac{1}{10^{12}},\quad\frac{1}{1000000},\quad\frac{1}{1250},\quad\frac{1}{1000}.$$

Thus dimension 7 is excluded and the proof is complete. □

4. Prescribed zeroes in dimensions 14 and 15

Theorem 4.1. *The Dittert conjecture holds in dimensions $n=14$ and $n=15$.*

We first record the exact finite calculation used in the proof. Put $N=n-2$ and

$$T_n(x)=\begin{pmatrix}0&a&b\mathbf{1}^T\\a&0&b\mathbf{1}^T\\b\mathbf{1}&b\mathbf{1}&cJ_N\end{pmatrix},\qquad a=1-Nx,\quad b=x,\quad c=\frac{1-2x}{N}.$$

Classifying permutations by the images of the first two rows gives

$$\begin{aligned}P_n(x):=\operatorname{per}T_n(x)&=N!a^2c^N+2N^2(N-1)!\,ab^2c^{N-1}\\&\quad+\big(N(N-1)\big)^2(N-2)!\,b^4c^{N-2}.\end{aligned}$$

Lemma 4.2. *Let*

$$q_n = \min_{0\le x\le 1/(n-2)} P_n(x).$$

For $n = 14, 15$,

$$q_n - \gamma_n - \frac{n^3 q_n^2}{4(1-\gamma_n)} > 0.$$

Proof. Appendix B gives the complete exact procedure and its rational output. The Sturm variation count shows that P_n' has exactly one root in the relevant compact interval. Rational interval Horner evaluation at the isolating interval and exact endpoint evaluation give

$$q_{14} > \frac{15781}{2000000000}, \qquad q_{15} > \frac{3001}{1000000000}.$$

Substitution into the displayed expression gives positive rational margins strictly greater than $\frac{1}{500000000}$ and $\frac{7}{1000000000}$, respectively. The expression is increasing throughout the rational interval containing q_n, as follows from the last cross-multiplication in that appendix. □

Proof of Theorem 4.1. Let A be a global maximizer. By Theorem 2.2, it is enough to consider the boundary. Write

$$\operatorname{per} A = \gamma_n - \delta.$$

Lemma 2.1 gives $0 < \delta \le \gamma_n$, and Corollary 2.4 supplies two zeroes in distinct rows and columns.

Set

$$t = \sqrt{\frac{n\delta}{1-\delta}}.$$

For $n = 14, 15$, $t < 1$. Kafidov's joint-deficit scaling lemma [15, Lemma 3.1] says that $(1-t)^{-1}A$ dominates a doubly stochastic matrix B. Domination preserves the two independent zeroes, which may be moved to $(1,1)$ and $(2,2)$.

Minc proved that the face of doubly stochastic matrices with these two prescribed zeroes has a minimizing symmetric matrix [19]. His exact later restatement displays $T_n(\alpha)$ and the cubic determining α [20, p. 153]. The printed formula for the corner entry has $n-1$ where its row sum forces $n-2$; with this typographical correction it is precisely the matrix above, and Minc's cubic is an exact factor of P_n'. Since every $T_n(x)$ lies in the face, Minc's theorem gives $\operatorname{per} B \ge q_n$. Therefore

$$\gamma_n - \delta = \operatorname{per} A \ge q_n(1-t)^n. \tag{19}$$

Bernoulli's inequality and $\delta \le \gamma_n$, with $z = \sqrt{\delta}$, give

$$\begin{aligned} q_n(1-t)^n - (\gamma_n - \delta) &\ge q_n - \gamma_n + z^2 - \frac{n q_n \sqrt{n}}{\sqrt{1-\gamma_n}} z \\ &\ge q_n - \gamma_n - \frac{n^3 q_n^2}{4(1-\gamma_n)} > 0 \end{aligned}$$

by Lemma 4.2. This is the strict reverse of (19), excluding a boundary maximizer. Thus $A = U_n$, including the claimed uniqueness. □

Corollary 4.3. *Together with Pang's and Kafidov's prior results, the Dittert conjecture holds for every* $n \ge 7$.

Limit of this argument. The same exact Sturm procedure gives a negative result for this particular method in dimension 13: at $\delta = \gamma_{13}/91$, a rational upper bound on q_{13} and a rational lower bound on the square root prove

$$q_{13}(1 - \sqrt{13\delta/(1-\delta)})^{13} < \gamma_{13} - \delta.$$

Thus merely removing the Bernoulli slack cannot settle dimension 13; Theorem 3.1 reaches it by a genuinely different spectral route. At this stage of the argument dimensions $5, 6$ remain; the uniform boundary proof below subsumes them and every dimension above them. The individual dimension arguments are components of the single all-dimensional theorem.

5. The order-four boundary

There is a useful probabilistic reformulation of the exact target. Put $p_{ij} = a_{ij}/n$, draw n independent pairs with law p, and let $\mathcal{R}$ and $\mathcal{C}$ be the events that all row labels and all column labels, respectively, occur. Direct counting gives

$$\Pr(\mathcal{R}) = \gamma_n R, \qquad \Pr(\mathcal{C}) = \gamma_n C, \qquad \Pr(\mathcal{R} \cap \mathcal{C}) = \gamma_n P.$$

Consequently

$$\Pr(\mathcal{R} \cup \mathcal{C}) = \gamma_n \Phi(A).$$

Thus Dittert's conjecture says exactly that the uniform law uniquely maximizes this row-or-column complete-coverage probability.

A sharp capacity reduction does not by itself settle any of the three remaining dimensions. Gurvits and Leake prove, without a total-support hypothesis, that a row-stochastic matrix with column-sum vector α entrywise dominates μD for some doubly stochastic D, where $\mu \geq 1 - \|\alpha - \mathbf{1}\|_1/2$ [11, Theorem 5.5]. Combining this with the exact stationarity identity

$$\alpha - \mathbf{1} = \frac{\Phi(A)}{R}(c - \mathbf{1})$$

and Minc's two-zero face constant yields a scalar lower bound for P. The rational countermodels in Appendix C show that this bound is insufficient in each of $n = 4, 5, 6$, even after retaining the common product deficit and the sharp product-to-L^1 relation. Therefore any successful continuation must use entrywise or multiplier information that these marginal data discard.

Proposition 5.1. *Up to row permutations, column permutations, and transpose, a dimension-four boundary maximizer has one of* 23 *possible zero-support types. Their numbers according to the number of zeroes* $2, \ldots, 8$ *are*

$$1,\ 3,\ 6,\ 5,\ 5,\ 2,\ 1.$$

In fact, no dimension-four boundary maximizer exists.

Proof. By Theorem 2.3, a boundary maximizer is fully indecomposable, and Corollary 2.4 shows that its zero graph contains two independent edges. Exhausting the 2^{16} zero masks subject to these two conditions and taking the indicated group orbits gives the displayed list. Appendix D prints all 23 orbit representatives and gives pseudocode for the exhaustion, so the classification does not rely on an external file.

Now let A be any boundary maximizer and move an independent zero pair to $(1,1), (2,2)$. Let $\mathcal{Z}$ be the stratum of matrices having exactly these two zeroes. If $B \in \mathcal{Z}$, then $(1-\varepsilon)A + \varepsilon B \in \mathcal{Z}$ for every $\varepsilon > 0$. Continuity therefore makes the supremum of Φ on $\mathcal{Z}$ equal to the global maximum. Hwang's attained-stratum averaging lemma [7, Lemma 3.4] supplies $Z_0 \in \mathcal{Z}$ attaining that supremum, with

equal rows $3,4$ and equal columns $3,4$. Consequently Z_0 is a global maximizer of the form

$$A=\begin{pmatrix} 0 & a & b & b \\ c & 0 & d & d \\ e & f & g & g \\ e & f & g & g \end{pmatrix}, \qquad a,b,c,d,e,f,g>0.$$

Put

$$a=x+\xi, \quad c=x-\xi, \quad b=y+\eta, \quad d=y-\eta,$$
$$e=w+\zeta, \quad f=w-\zeta,$$

so $|\xi|<x$, $|\eta|<y$, $|\zeta|<w$. If $F=\Phi(A)$ and F_0 is its value at $\xi=\eta=\zeta=0$, exact expansion gives

$$\begin{aligned} F-F_0 = 2[&-D\xi^2-K_w\xi\eta+K_y\xi\zeta-M_w\eta^2+4gx\eta\zeta \\ &-M_y\zeta^2-2\eta^2\zeta^2], \end{aligned} \tag{D4.1}$$

where

$$D=3g^2+4g(w+y)+2(w^2+y^2),$$
$$K_w=8g^2+12gw+8w^2, \quad K_y=8g^2+12gy+8y^2,$$
$$M_w=8g^2+16gw+6w^2, \quad M_y=8g^2+16gy+6y^2.$$

Odd stationarity first gives

$$\xi=\frac{-K_w\eta+K_y\zeta}{2D}.$$

After substitution, (D4.1) becomes

$$-\alpha\eta^2+2\beta\eta\zeta-\delta\zeta^2-4\eta^2\zeta^2, \tag{D4.2}$$

where

$$\alpha=2M_w-\frac{K_w^2}{2D}, \quad \delta=2M_y-\frac{K_y^2}{2D}, \quad \beta=4gx-\frac{K_wK_y}{2D}.$$

At a nonzero odd stationary point,

$$\alpha\eta=\beta\zeta-4\eta\zeta^2, \qquad \delta\zeta=\beta\eta-4\eta^2\zeta, \qquad \alpha\eta^2=\delta\zeta^2. \tag{D4.3}$$

The determinant of the (η,ζ)-Hessian there is $64\alpha\eta^2$. If $\eta\zeta=0$, (D4.2)–(D4.3) show that the fixed midpoint has the same objective, but no relation between the independent positive parameters y and w is needed: the one-sided derivative identity (D4.4) below directly excludes this case. We may therefore analyze the remaining case $\eta\zeta\neq0$. Local maximality gives $\alpha,\delta\geq0$. Their simultaneous vanishing is impossible: after normalizing $g=1$, put $s=y+w$, $p=yw$. The difference of their cleared numerators is

$$(y-w)\{-7s-4(s^2-2p)+2s(s^2-2p)\}.$$

The case $y=w$ makes $\alpha>0$. Otherwise simultaneous vanishing gives $2(s-2)(s^2-2p)=7s$. Since $p>0$, this forces $s>3$, while their cleared sum is a positive multiple of $N(s)/(s-2)^2$, where

$$\begin{aligned} N(t+3) = &\,4t^6+68t^5+452t^4+1444t^3 \\ &+2145t^2+1149t+158>0. \end{aligned}$$

Thus $\alpha,\delta>0$.

For $(i,j)=(1,1),(2,2)$, let Z_{ij} be the difference between the missing-entry derivative and the common positive-entry derivative. Direct differentiation gives

$$Z_{11}+Z_{22}=4g(gx-4\eta\zeta). \tag{D4.4}$$

Indeed the column-product derivative terms cancel in the sum, while the four permanental cofactors are

$$4dfg, \quad 2cg^2 + 4deg, \quad 4beg, \quad 2ag^2 + 4bfg,$$

which give (D4.4) after substituting the odd coordinates. If $\eta\zeta \le 0$, this is positive and one zero violates Hwang's one-sided condition. Suppose $p_0 = \eta\zeta > 0$, and put

$$L = \frac{K_w K_y}{2D}, \quad T = L + \sqrt{\alpha\delta}, \quad m = \min\left\{ y^2\sqrt{\frac{\alpha}{\delta}}, w^2\sqrt{\frac{\delta}{\alpha}} \right\}.$$

Equations (D4.3) and entry positivity give

$$p_0 = gx - \frac{T}{4}, \qquad p_0 < m. \tag{D4.5}$$

The remaining exact coefficient inequality is

$$T \ge 12m. \tag{D4.6}$$

For completeness, suppose $w \ge y$, normalize $y = 1$, and write $t = g/y$, $q = w/y$; the other case is symmetric. One checks that the first term defines m. If $\delta \ge 12$, (D4.6) is immediate. Otherwise, clearing denominators first forces $0 < t < 5/16$; $\alpha > 0$ then forces $1 \le q < 3$. Squaring (D4.6) is equivalent to

$$L^2\delta - (12 - \delta)^2\alpha \ge 0. \tag{D4.7}$$

After its positive denominator is cleared and $t = 5u/16$, $q = 1 + 2v$, all 77 exact tensor-product Bernstein coefficients of bidegree $(10, 6)$ are nonnegative; the relevant positivity principle is the convex-hull property of the Bernstein basis [10]. Only the coefficient at $(0, 0)$ is zero. Appendix E derives the minimum selector and the two rectangle reductions, defines the Bernstein conversion, and prints all 77 scaled integer coefficients. Hence (D4.6) holds, strictly here.

Finally, if (D4.4) were nonpositive, (D4.5) would give $p_0 \ge gx/4$, then $gx \ge T/3$, and consequently $p_0 \ge T/12 \ge m$, contradicting (D4.5). Thus one of the two missing entries always violates the one-sided multiplier condition. □

Corollary 5.2. *The Dittert conjecture holds in dimension* 4.

Proof. There is no boundary maximizer by Proposition 5.1. Hwang's positive-support theorem makes U_4 the unique positive maximizer. □

6. The uniform boundary argument

Proposition 6.1. *The Dittert conjecture holds in every dimension* $n \ge 5$.

Proof. Put $m = n - 2$, so $m \ge 3$. By Corollary 2.4, every boundary maximizer has two zeroes in distinct rows and columns. Move them to $(1, 1), (2, 2)$, and let $\mathcal{Z}$ be the stratum having exactly these zeroes. Continuity under convex mixing makes the supremum on $\mathcal{Z}$ equal to the global maximum. Hwang's attained-stratum lemma [7, Lemma 3.4] therefore supplies a global maximizer in $\mathcal{Z}$ of the form

$$A = \begin{pmatrix} 0 & a & b & \cdots & b \\ c & 0 & d & \cdots & d \\ e & f & g & \cdots & g \\ \vdots & \vdots & \vdots & & \vdots \\ e & f & g & \cdots & g \end{pmatrix}, \qquad a, b, c, d, e, f, g > 0, \tag{D56.1}$$

where the final row and column classes have size m.

Write

$$a = x + \xi, \quad c = x - \xi, \quad b = y + \eta, \quad d = y - \eta, \quad e = w + \zeta, \quad f = w - \zeta.$$

The three odd perturbations preserve total mass, and

$$|\xi| < x, \qquad |\eta| < y, \qquad |\zeta| < w. \tag{D56.2}$$

The permanent is

$$\operatorname{per} A = m!\{acg^m + m(ade + bcf)g^{m-1} + m(m-1)bdefg^{m-2}\}. \tag{D56.3}$$

Put

$$u = (2w + mg)^m, \qquad v = (2y + mg)^m, \qquad D = u + v - m!g^m,$$
$$K_w = m(u - m!g^{m-1}w), \qquad K_y = m(v - m!g^{m-1}y),$$
$$M_w = m^2u - m(m-1)m!g^{m-2}w^2,$$
$$M_y = m^2v - m(m-1)m!g^{m-2}y^2, \qquad N = m(m-1)m!g^{m-2}.$$

Since $m^m > m!$, the binomial expansion gives $D, K_w, K_y > 0$. Expanding the two marginal products and (D56.3) gives

$$\begin{aligned}\Phi(A) - \Phi(A)|_{\xi=\eta=\zeta=0} = & -D\xi^2 - 2K_w\xi\eta + 2K_y\xi\zeta - M_w\eta^2 \\ & + 2mm!g^{m-1}x\eta\zeta - M_y\zeta^2 - N\eta^2\zeta^2.\end{aligned} \tag{D56.4}$$

Odd stationarity first gives $\xi = (-K_w\eta + K_y\zeta)/D$. After substitution, (D56.4) is

$$-\alpha\eta^2 + 2\beta\eta\zeta - \delta\zeta^2 - N\eta^2\zeta^2, \tag{D56.5}$$

where

$$\alpha = M_w - \frac{K_w^2}{D}, \quad \delta = M_y - \frac{K_y^2}{D}, \quad \beta = mm!g^{m-1}x - \frac{K_wK_y}{D}. \tag{D56.6}$$

Its stationary equations imply

$$\alpha\eta^2 = \delta\zeta^2. \tag{D56.7}$$

Let Z_{11}, Z_{22} denote the derivative at a missing entry minus the common derivative at a positive entry. Hwang's multiplier condition [7, Lemma 3.6] requires both quantities to be nonpositive. Comparing with entries $(1,2),(2,1)$ cancels the marginal-product derivatives. The four permanental cofactors give

$$Z_{11} + Z_{22} = 2m!g^{m-1}(gx - 2m\eta\zeta). \tag{D56.8}$$

This is positive if $\eta\zeta \le 0$, already contradicting the two one-sided multiplier conditions. Suppose $p = \eta\zeta > 0$. The determinant of the (η,ζ)-Hessian of (D56.5), simplified with its stationary equations, is $16N\alpha\eta^2$. Local maximality gives $\alpha \ge 0$. The strict lower bound on δ below and (D56.7) then give $\alpha > 0$.

Put

$$L = \frac{K_wK_y}{D}, \qquad T = L + \sqrt{\alpha\delta}.$$

The stationary equations and (D56.6) give

$$p = \frac{gx}{m-1} - \frac{T}{N}, \qquad \frac{\eta}{\zeta} = \sqrt{\frac{\delta}{\alpha}}. \tag{D56.9}$$

Entry positivity therefore implies

$$p < p_* := \min\left\{y^2\sqrt{\frac{\alpha}{\delta}}, w^2\sqrt{\frac{\delta}{\alpha}}\right\}. \tag{D56.10}$$

If (D56.8) were nonpositive, then $p \ge gx/(2m)$; combining this with (D56.9) gives

$$p \ge \frac{T}{m(m+1)m!g^{m-2}}. \tag{D56.11}$$

It remains to contradict (D56.10)–(D56.11). By interchanging y, w, assume $w \ge y$, normalize $y = 1$, and put $t = g/y,\ q = w/y$. We prove, uniformly for $m \ge 3$,

$$\delta \ge s_m := m(m+1)m!t^{m-2}, \qquad q^2\delta \ge \alpha. \tag{D56.12}$$

Set

$$U=(2q+mt)^m,\quad V=(2+mt)^m,\quad D=U+V-m!t^m,$$
$$A_m=M_wD-K_w^2,\qquad \Delta_m=M_yD-K_y^2.$$

Only D depends on q in $\delta=M_y-K_y^2/D$, and D is increasing, so δ is increasing. Put $b=m!t^{m-2}$. At $q=1$, set $P=(2+mt)^m$. Then $D=2P-bt^2,\ K_y=m(P-bt)$, and $M_y=m^2P-m(m-1)b$. Consequently

$$\begin{aligned}\frac{\Delta_m-s_mD}{m^2}&=(P-2b)(2P-bt^2)-(P-bt)^2\\&=P^2-b(t^2-2t+4)P+b^2t^2,\end{aligned}$$

which is equivalently

$$\Delta_m-s_mD=m^2b^2\{z^2-((t-1)^2+3)z+t^2\},\qquad z=\frac{(2+mt)^m}{b}. \tag{D56.13}$$

The t^m and t^{m-2} binomial terms imply

$$z>t^2+\frac{2m^{m-2}}{(m-2)!}\ge t^2+6>(t-1)^2+3.$$

The middle quantity is 6 at $m=3$; its successive ratio is $((m+1)/m)^{m-2}(m+1)/(m-1)>1$. Hence the brace in (D56.13) is positive, proving the first inequality.

For the second inequality put $c=m!t^{m-1}$. The lower terms in $q^2M_y-M_w$ cancel, leaving $m^2(q^2V-U)$. Expanding $(q^2M_y-M_w)D+K_w^2-q^2K_y^2$, with $K_w=m(U-cq), K_y=m(V-c), D=U+V-ct$, and cancelling the square terms gives

$$\frac{q^2\Delta_m-A_m}{m^2}=UV(q^2-1)+m!t^{m-1}\{U(t-2q)+q^2V(2-t)\}. \tag{D56.14}$$

Put $q=1+v,\ W=2+mt$, and

$$U=\sum_{k=0}^{m}u_kv^k,\qquad u_k=\binom{m}{k}2^kW^{m-k},$$

with $u_k=0$ outside its displayed range. The coefficient C_k of v^k on the right of (D56.14) is

$$\begin{aligned}C_k&=W^m(2u_{k-1}+u_{k-2})+m!t^{m-1}H_k,\\H_k&=(t-2)u_k-2u_{k-1}\\&\quad+(2-t)W^m(\mathbf{1}_{k=0}+2\mathbf{1}_{k=1}+\mathbf{1}_{k=2}).\end{aligned} \tag{D56.15}$$

Here $C_0=0$. For $k=1$,

$$C_1=2W^{m-1}\left[W^{m+1}-m!t^{m-1}\{mt^2-2(m-1)t+2(m-1)\}\right].$$

Only degrees $m-1, m+1$ in the bracket can be negative. Their positive coefficients $2(m+1)m^m, m^{m+1}$ exceed $2(m-1)m!, m\,m!$, respectively.

For $k=2$,

$$\begin{aligned}C_2&=W^{m-2}\{W^{m+1}(4m+W)+m!t^{m-1}R_m(t)\},\\R_m(t)&=-4(m-1)(m+2)-2(m-1)(m-2)t\\&\quad+2m(m-2)t^2-m^2t^3.\end{aligned}$$

Only degrees $m-1, m, m+2$ can be negative. The first two are dominated by $4mW^{m+1}$, since

$$8m(m+1)m^m>4(m-1)(m+2)m!,\quad 8m(m+1)m^m>2(m-1)(m-2)m!,$$

and $m^{m+2}>m^2m!$ handles the last.

For $3 \le k \le m$,

$$H_k = -\frac{2^k(m+1)}{m-k+1}\binom{m}{k}W^{m-k}\{(k-1)t+2\}.$$

After factoring $2^{k-2}W^{m-k}$, the remaining polynomial is

$$4\binom{m}{k-1}W^{m+1} + \binom{m}{k-2}W^{m+2} - \frac{4m!(m+1)}{m-k+1}\binom{m}{k}t^{m-1}\{(k-1)t+2\}.$$

Only degrees $m-1, m$ can be negative. The coefficient of the first positive term at both degrees is $8\binom{m}{k-1}(m+1)m^m$. Using $\binom{m}{k}/(m-k+1) = \binom{m}{k-1}/k$, the negative coefficients are

$$\frac{8m!(m+1)}{k}\binom{m}{k-1}, \qquad \frac{4m!(m+1)(k-1)}{k}\binom{m}{k-1},$$

both smaller because $m^m > m!$. Finally,

$$C_{m+1} = 2^{m-1}\{W^m(4+mW) - 4m!t^{m-1}\} >_{\text{coeff}} 0, \quad C_{m+2} = 2^m W^m >_{\text{coeff}} 0;$$

at the only threatened degree of C_{m+1}, $4W^m$ contributes $8m^m > 4m!$. Thus every nonconstant coefficient in (D56.14) is positive, proving the second inequality.

Appendix F records a direct symbolic-expansion scheme for the permanent, odd expansion, cofactor sum, and coefficient inequalities. It is included only as transparent bookkeeping for the lowest multiplicities $m = 3, 4$; the all-m proof is the coefficient argument above.

The second inequality in (D56.12) selects the first term in (D56.10). Restoring the normalization gives $\delta \ge m(m+1)m!g^{m-2}y^2$. The first inequality, together with $L > 0$, therefore gives

$$T \ge \sqrt{\alpha\delta} = \delta\sqrt{\alpha/\delta} \ge m(m+1)m!g^{m-2}p_*.$$

This contradicts (D56.10)–(D56.11). Thus (D56.8) is always positive, so (D56.1) cannot be a maximizer. The stratum bridge excludes every boundary maximizer, and Hwang's positive-support theorem identifies the unique maximizer as U_n. □

7. Completion of the proof

Proof of Theorems 1.3 and 1.2. Proposition 5.1 excludes boundary maximizers in order four, and Proposition 6.1 excludes them in every order $n \ge 5$. This proves Theorem 1.3. Theorem 2.2 then identifies the unique maximizer as U_n for every $n \ge 4$. Sinkhorn's theorem gives $n = 2$, and Hwang's theorem gives $n = 3$ [22, 14]. Hence Conjecture 1.1 holds in every dimension, proving Theorem 1.2. □

Appendix A. Exact rational verification of the spectral alternatives

This appendix specifies the finite arithmetic used in Theorem 3.1. It is included to make the argument reproducible from this paper alone. The Sturm-sequence and exact root-isolation procedures are standard; see [1]. Every input is one of the rational numbers printed in the proof, and every comparison is performed after cross-multiplication over $\mathbb{Z}$.

For a polynomial $p(x) = \sum_{i=0}^{d} p_i x^i$ and a rational interval $I = [a, b]$, define interval Horner evaluation recursively by

$$H_d = [p_d, p_d], \qquad H_i = IH_{i+1} + [p_i, p_i],$$

where the product of two intervals is the interval spanned by the four endpoint products. Thus $p(I) \subseteq H_0$, with no rounding. For a rational symmetric matrix M, exact LDL^T elimination uses

$$d_i = M_{ii} - \sum_{k<i} l_{ik}^2 d_k, \qquad l_{ji} = \frac{M_{ji} - \sum_{k<i} l_{jk} l_{ik} d_k}{d_i}.$$

All pivots $d_i > 0$ certify positive definiteness.

The finite procedure is the following.

(1) Evaluate every displayed AM–GM marginal inequality and every block-permanent inequality as a rational number.
(2) For every block order m, enumerate the pairs $0 \le i, j \le m$ with $i + j > m$ and check the displayed Hall-loss inequality.
(3) Form the weighted path matrix from the displayed H_ℓ and marginal bounds and apply the exact LDL^T recursion above.
(4) For $n = 8$, subdivide $[0, 491/10^4]$ into 16 equal rational intervals and apply interval Horner evaluation to each of the four degree-eight polynomials in (14).
(5) For $n = 7$, subdivide $[0, 783/10^4]$ into 4096 equal rational intervals. On each interval and for each $\ell = 1, 2, 3$, choose the least $B \in 10^{-8}\mathbb{Z}$ for which the first polynomial in (18) has positive interval-Horner lower endpoint; then check the second polynomial in (18) with that same B. Finally apply interval Horner evaluation to the cubic spectral comparison following (18).

The following table gives strict rational lower bounds for the smallest output in every finite family of comparisons. A dash means that the corresponding dimension uses no check of that type. The headings "local 1" and "local 2" refer, in order, to the additional Hall/leakage or Hall/polynomial tests described in the proof.

(A.1)

n	marginal or a_n	local 1	local 2	block	path pivot	spectral
7	$3/10^5$	$1/10^3$	$1/10^{12}$	$1/10^6$	$1/1250$	$1/10^3$
8	$1/200000$	$1/10^3$	$1/20000$	–	$1/500$	$3/10^4$
9	$1/500000$	$3/10^6$	–	$1/125000$	$1/5000$	$1/25000$
10	$19/10^9$	–	–	$9/10^7$	–	$1/10^4$
11	$1/(4 \cdot 10^7)$	–	–	$1/250000$	–	$3/50000$
12	$1/10^6$	–	–	$1/10^6$	–	$1/2500$
13	$9/10^7$	–	–	$1/10^6$	–	$1/2000$

Each entry is a deliberately shortened lower bound, not a decimal approximation to the actual margin.

Appendix B. The prescribed-two-zero Sturm calculation

Let P_n be the polynomial displayed before Lemma 4.2. Starting with

$$S_0 = P_n', \qquad S_1 = P_n'', \qquad S_{k+1} = -\operatorname{rem}(S_{k-1}, S_k),$$

delete zero terminal terms to obtain the Sturm sequence. At a rational number x, let $V(x)$ be the number of sign changes in $(S_0(x), S_1(x), \ldots)$, after zero entries are omitted. The number of roots of P_n' in (a, b) is $V(a) - V(b)$. Repeated rational bisection isolates every such root, after which the interval-Horner rule of Appendix A bounds P_n there.

The complete output needed in Lemma 4.2 is

(B.1)

n	$\deg P_n$	critical-point isolating interval	# roots	q_n lower bound	$f_n(q_n)$ lower bound
14	14	$[7647323/10^8,\ 1911831/25000000]$	1	$15781/(2\cdot 10^9)$	$1/500000000$
15	15	$[3553391/50000000,\ 7106783/10^8]$	1	$3001/10^9$	$7/10^9$

where

$$f_n(q) = q - \gamma_n - \frac{n^3 q^2}{4(1-\gamma_n)}.$$

Exact endpoint evaluation gives

(B.2)

n	$P_n(0)$	$P_n(1/(n-2))$
14	$1925/35831808$	$206787109375/25999348907114496$
15	$479001600/23298085122481$	$1639976960656103731200/54280077037437051277159536 1$

and both endpoint values exceed the corresponding lower bound in (B.1). Interval Horner evaluation on the two isolating intervals gives the same strict lower bounds. Hence the table covers every possible global minimum.

Finally,

$$f_n'(q) = 1 - \frac{n^3 q}{2(1-\gamma_n)}.$$

Substituting the endpoint upper bound $q \le P_n(0)$ makes this positive for $n = 14, 15$, again by integer cross-multiplication. Therefore the lower bounds for $f_n(q_n)$ in (B.1) follow from the lower bounds for q_n.

For the stated limitation at $n = 13$, take

$$\delta = \frac{\gamma_{13}}{91} = \frac{68428800}{302875106592253}, \quad x_0 = \frac{82762562}{10^9}, \quad t_- = \frac{856898537}{5\cdot 10^{11}}.$$

Then $t_-^2 \le 13\delta/(1-\delta)$, and exact evaluation gives

$$P_{13}(x_0)(1-t_-)^{13} - (\gamma_{13} - \delta) < -\frac{9}{10^8}. \tag{B.3}$$

Thus the scalar comparison really does fail in dimension 13.

Appendix C. Capacity-route countermodels

The diagnostic in the dimension-four discussion is also completely rational. For the data (s, d, x) in the table, put

$$\rho = 1 - \left(1 - \frac{d}{s}\right)^s \left(1 + \frac{d}{n-s}\right)^{n-s},$$
$$\mu = 1 - \frac{(2-\gamma_n)d}{1-\rho}, \qquad q^+ = P_n(x),$$
$$M = q^+(1-\rho)\mu^n - (\gamma_n - 2\rho).$$

The following strict bounds follow by direct cross-multiplication.

(C.1)

n	s	d	x	ρ upper	q^+ upper	M upper
4	2	$147/1000$	$3793/12500$	$11/1000$	$103/1000$	$-45/1000$
5	2	$101/1000$	$4723/20000$	$43/10000$	$41/1000$	$-16/1000$
6	3	$69/1000$	$963/5000$	$16/10000$	$17/1000$	$-5/1000$

Since $M < 0$ in every row, even the strengthened scalar capacity route does not reach the required permanent lower bound.

Appendix D. Complete order-four support enumeration

Encode a zero set by a 16-bit mask. A mask is admissible if

(1) it contains two entries in distinct rows and distinct columns; and
(2) for no nonempty proper $I \subset \{1,2,3,4\}$ does it contain the entire rectangle $I \times J$ with $|J| = 4 - |I|$.

The second condition is precisely the zero-pattern test for full indecomposability. For every admissible mask, take the lexicographically least image under independent row and column permutations and transpose. The following pseudocode is an exhaustive specification:

$$
\begin{aligned}
&\mathcal{C} \leftarrow \varnothing;\\
&\textbf{for } z = 1, \dots, 2^{16} - 1 :\\
&\quad \textbf{if } z \text{ satisfies (1) and (2):}\\
&\qquad c \leftarrow \min\{\tau(\sigma_r z \sigma_c) : \sigma_r, \sigma_c \in S_4,\ \tau \in \{1, \text{transpose}\}\};\\
&\qquad \mathcal{C} \leftarrow \mathcal{C} \cup \{c\}.
\end{aligned}
\tag{D.1}
$$

Writing ij for the zero entry (i, j), the resulting 23 representatives are:

#	$\lvert Z\rvert$	Z	#	$\lvert Z\rvert$	Z
1	2	12, 21	13	5	12, 14, 22, 23, 31
2	3	11, 12, 21	14	5	13, 14, 22, 31, 41
3	3	12, 13, 21	15	5	11, 14, 23, 32, 41
4	3	13, 22, 31	16	6	12, 13, 21, 23, 31, 32
5	4	11, 13, 21, 22	17	6	12, 14, 21, 23, 31, 32
6	4	13, 14, 21, 22	18	6	12, 14, 22, 23, 31, 41
7	4	12, 13, 21, 31	19	6	12, 14, 21, 23, 32, 41
8	4	11, 13, 22, 31	20	6	13, 14, 21, 23, 32, 41
9	4	13, 14, 22, 31	21	7	13, 14, 22, 23, 31, 32, 41
10	4	14, 23, 32, 41	22	7	13, 14, 22, 24, 32, 33, 41
11	5	12, 13, 21, 22, 31	23	8	13, 14, 22, 24, 31, 33, 41, 42
12	5	13, 14, 21, 22, 31			

(D.2)

Counting by $|Z| = 2, \dots, 8$ gives $(1, 3, 6, 5, 5, 2, 1)$, as asserted.

Appendix E. The order-four Bernstein certificate

We give the exact arithmetic suppressed in the transition from (D4.5) to (D4.7). After $y = 1$, $t = g/y$, and $q = w/y$, write

$$
\begin{aligned}
D &= 3t^2 + 4tq + 4t + 2q^2 + 2,\\
K_w &= 8t^2 + 12tq + 8q^2, \qquad K_y = 8t^2 + 12t + 8,\\
M_w &= 8t^2 + 16tq + 6q^2, \qquad M_y = 8t^2 + 16t + 6,\\
A &= 2M_w D - \tfrac{1}{2}K_w^2, \qquad \Delta = 2M_y D - \tfrac{1}{2}K_y^2.
\end{aligned}
$$

Thus $\alpha = A/D$, $\delta = \Delta/D$, and $L = K_w K_y/(2D)$.

First, the numerator

$$S(t, q) = q^2\Delta - A$$

satisfies, on writing $q = 1 + v$,

$$
\begin{aligned}
S(t, 1+v) = {} & (64v + 160v^2 + 128v^3 + 32v^4)\\
& + t(224v + 448v^2 + 288v^3 + 64v^4)\\
& + t^2(384v + 576v^2 + 256v^3 + 32v^4)\\
& + t^3(256v + 256v^2 + 64v^3) + t^4(32v + 16v^2).
\end{aligned}
\tag{E.1}
$$

Hence $q^2\delta \geq \alpha$ for $q \geq 1$, so the first term is the minimum in (D4.5).

Next put

$$\begin{aligned} j(t,q) &= (\Delta - 12D)/8 \\ &= -4 - 4t + 8tq^2 + 3t^2 + 16t^2q + 4t^2q^2 + 8t^3 + 8t^3q + 2t^4. \end{aligned}$$

It is increasing in $q \geq 1$, while

$$j(t,1) = 2t^4 + 16t^3 + 23t^2 + 4t - 4, \qquad j(5/16,1) = 113/32768 > 0.$$

Thus $\delta < 12$, equivalently $j(t,q) < 0$, forces $t < 5/16$. Finally let $i(t,q) = -A/4$. Its derivative with respect to t has strictly negative coefficients, and

$$i(5/16, 3+v) = \frac{2767}{16384} + \frac{27225}{256}v + \frac{21897}{256}v^2 + \frac{91}{4}v^3 + 2v^4 > 0. \tag{E.2}$$

Since $\alpha > 0$ means $i < 0$, (E.2) forces $q < 3$.

Define the polynomial h by

$$\frac{1}{4}K_w^2 K_y^2 \Delta - (12D - \Delta)^2 A = 1024D\, h(t,q). \tag{E.3}$$

Because $D > 0$, inequality (D4.7) is equivalent to $h \geq 0$. On the rectangle $t = 5u/16,\ q = 1 + 2v$, write

$$h(5u/16, 1+2v) = \sum_{r=0}^{10}\sum_{s=0}^{6} b_{rs}\binom{10}{r}u^r(1-u)^{10-r}\binom{6}{s}v^s(1-v)^{6-s}. \tag{E.4}$$

For compactness, the table gives the exact integers

$$\widehat{b}_{rs} = 16^{10}\binom{10}{r}\binom{6}{s}b_{rs}.$$

(E.5a)

$r\backslash s$	0	1	2	3
0	0	52776558133248	527765581332480	2005509209063424
1	10995116277760	755914244096000	6797730638725120	24744509183098880
2	131597797949440	4871953202544640	39783766666772480	138818668766167040
3	684338614108160	18402251773050880	138255740824453120	463593245528555520
4	2037861989744640	44889870077788160	314269358547271680	1015984788235878400
5	3840421773967360	73771295038242816	486571381652193280	1521712544442482688
6	4759162783989760	82707597515161600	518563451833221120	1573843691725127680
7	3883644615229440	62501158991626240	375187130506772480	1108144168546467840
8	2014444474940160	30493375333793280	176248115411877120	507813903405982720
9	603126347120640	8680816622338560	48525937286791680	136667246148121600
10	79459459141860	1095864216308568	5945463454398300	16395153287757264

(E.5b)

$r\backslash s$	4	5	6
0	3588805953060864	3008263813595136	949978046398464
1	43238294762291200	35720384007372800	11173786917273600
2	237112774906347520	193664284444590080	60436718205009920
3	775102262820208640	627284432293724160	196151556377149440
4	1665176387746529280	1337219171766763520	419932056877793280
5	2448320986913701888	1952660699807219712	616306674696716288
6	2488951360659456000	1972460755271680000	625677506791014400
7	1724493900574392320	1358340519418593280	432834714309918720
8	778409822206915840	609511543169768960	194986892197712640
9	206526568418452480	160778523305208320	51607187621460480
10	24442862602674524	18920540600095576	6090767843652324

All 77 entries in (E.5a)–(E.5b) are nonnegative and only $\widehat{b}_{00}$ vanishes. Since the tensor-product Bernstein basis is nonnegative on $[0,1]^2$, (E.4)–(E.5b) prove $h \geq 0$, with strict positivity in the case used in the proof.

Appendix F. Bookkeeping for the uniform multiplicity identities

The universal argument following (D56.12) is symbolic and does not depend on finite testing. For transparency, its lowest-multiplicity expansion can be reproduced by the following finite algebraic recipe:

(1) Form the seven-variable block matrix (D56.1), retaining the last row and column multiplicity as the formal integer m.
(2) Enumerate the three possible assignment types of the first two rows: both use the two exceptional columns, exactly one does, or neither does. Their multiplicities give the three terms of (D56.3).
(3) Substitute the odd coordinates, expand the two marginal products and (D56.3), and collect in ξ, η, ζ. This gives (D56.4).
(4) Form D, K_w, K_y, M_w, M_y, N, eliminate ξ, and collect the numerator of each assertion in (D56.12) first in $q-1$ and then in t.

For the two lowest multiplicities, the exact coefficient bookkeeping is:

(F.1)

m	n	# lower-numerator terms for δ	least positive coefficient	# selector terms, least coefficient
3	5	7	576	$(29, 576)$
4	6	9	4096	$(44, 4096)$

These rows are consistency checks on the displayed identities, not a substitute for the coefficientwise proof for arbitrary m.

Statement on AI-assisted discovery and human verification

The TARS agent system assisted with exploratory derivations with an autonomous mathematical search. X. Dai completed proof reconstruction, supplemented omitted details, literature review and manuscript writing, and manually verified all mathematical results.

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